Supplementary Experiments

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In this supplementary material, we report extra numerical results that are not included in our formal manuscript. We refer to the algorithm in [1] as RDS, and denote our proposed algorithm as MDS.

Following matrices are used in [1] to illustrate the stability of their algorithm:

$$A = \begin{bmatrix} a & bk & a_{1,3} & a_{1,4} \\ -b/k & a & a_{2,3} & a_{2,4} \\ 0 & 0 & a + r_1 g & (b + r_2 g)k \\ 0 & 0 & -(b + r_2 g)/k & a + r_1 g \end{bmatrix},$$

in which parameters k and g are used to change the gap between eigenvalues while other variables are drawn from the standard normal distribution. We reproduced the experiments in [1] by examining the relative error of eigenvalues after swapping by two algorithms. Let $\check{A}$ denote the matrix A after swapping. Let λ_1 and λ_2 denote the eigenvalues associated with $A_{1,1}$ and $A_{2,2}$, respectively. And let $\check{\lambda}_1$ and $\check{\lambda}_2$ denote the eigenvalues associated with $\check{A}_{2,2}$ and $\check{A}_{1,1}$, respectively. All experiment settings are kept consistent with those described in [1]. Figures 1 and 2 show the results.

As shown in Figure 1, both algorithms suffer a severe loss of precision for λ_1 in parameter domain $g \geq 10^4$, $k \geq 10^4$ or $k \leq 10^{-4}$. In the worst case, the relative error of λ_1 for both algorithms is of the order 10^4. However, when $g \leq 10^{-4}$, $k \leq 10^{-4}$, the relative error of λ_1 for RDS is clearly larger than the error for MDS. Figure 2 illustrates that the significant instability of λ_2 only appears in the parameter domain $g \leq 10^{-4}$, $k \leq 10^{-4}$ for RDS. Indeed, even in the worst case, the relative error of λ_2 for MDS is merely of the order 10^{-10}. This is remarkable smaller than the error for RDS, which is of the order 10^{-2}. Thus, MDS generally outperforms RDS in these numerical experiments.

References

[1] D. Camps, N. Mastronardi, R. Vandebril, and P. Van Dooren. Swapping 2 × 2 blocks in the Schur and generalized Schur form. *J. Comput. Appl. Math.*, 373:112274, 2020.

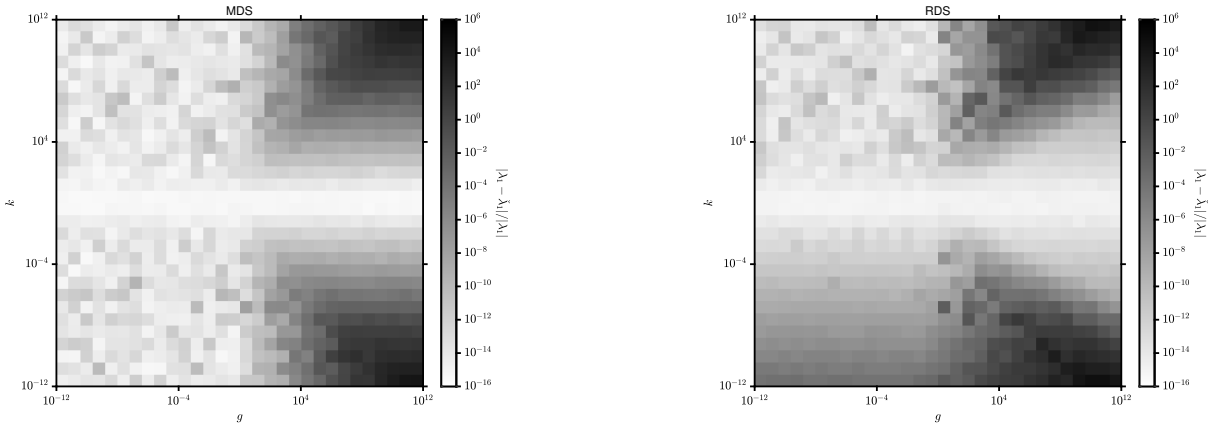


Figure 1: Relative error of λ_1.

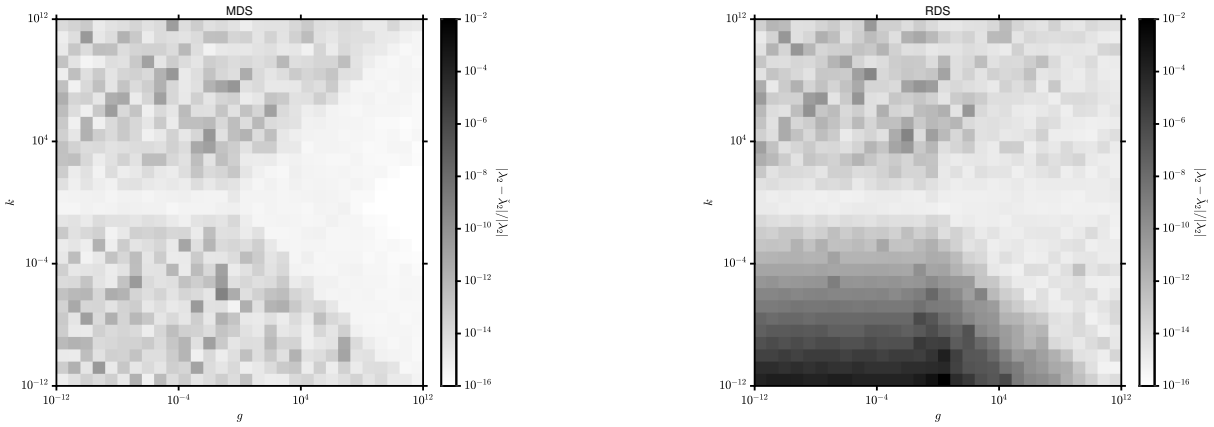


Figure 2: Relative error of λ_2.